

Aaryan Patel

SYSTEMS & SOFTWARE ENGINEER

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Computer Science senior graduating December 2026 and Industrial IoT/Data Engineering intern with more than one year of experience building production data workflows, operational analytics, fault-tested distributed systems, and ML infrastructure. C++17, Python, and SQL work spans Raft/MVCC, vectorized query execution, paged-KV scheduling, SEC data services, predictive maintenance, and Power BI.

TECHNICAL SKILLS

Languages: C++17, Python, SQL, TypeScript/JavaScript, R

Systems: Distributed systems, Raft, MVCC, WAL/LSM trees, concurrency, vectorized execution, fault injection, linearizability

ML & Data: PyTorch, HNSW/FAISS, LLM inference, ETL, data modeling, time-series and sensor analytics, signal processing

Tools: Linux, Git/GitHub Actions, Docker, Power BI, Microsoft M1 ERP, PLC and Industrial IoT integration

EXPERIENCE

MP Equipment

Gainesville, Georgia

Industrial IoT & Data Engineer Intern

May 2025 - Present

Computer Science Intern -> Industrial IoT & Data Engineering Intern -> current role

- Build Python and SQL pipelines that combine IoT sensor streams, equipment alarm history, and M1 ERP records; standardize assets, measurements, missing values, and outlier handling for reliable reporting.
- Develop predictive-maintenance workflows from vibration, temperature, humidity, and machine-state signals to surface abnormal behavior and support maintenance decisions.
- Design Power BI dashboards for OEE, downtime, on-time delivery, vendor performance, production KPIs, and equipment health used by engineering and operations.
- Integrate PLC-connected machinery with real-time monitoring and automate recurring operational reports.

Georgia State University - Department of Computer Science

Atlanta, Georgia

Undergraduate Research Assistant

Aug 2022 - May 2023

- Analyzed Reddit social-network data for patterns in group relationships and maintained research data models.
- Built date-range controls and public-facing views for machine-learning and data-mining research data.

SELECTED ENGINEERING PROJECTS

Raft + MVCC Database Lab | C++17

- Built five-node Raft consensus, serializable MVCC/OCC, and a linearizability checker; 598 assertions across seeded faults, 11-logical-tick failover, and minority leaders prevented from committing.

Paged LLM Inference Scheduler | Python, PyTorch

- Implemented paged KV caching, continuous batching, chunked prefill, and prefix caching; 90 tests, 94% vs. 21% KV utilization, 2.9x sequential throughput, and 5.5x better median TTFT in a CPU-scale benchmark.

SEC EDGAR MCP Server | Python, MCP, XBRL

- Built filing search, identity resolution, bounded text windows, XBRL discovery, global request pacing, and host-aware caching; 32 offline tests and a measured 138x warm 10-K read improvement.

Vectorized Columnar Engine | C++17

- Built null-aware vectorized filters, joins, aggregation, and TPC-H Q1; measured 53.1M rows/second with 6,288 assertions and 1.94x scalar-filter throughput.

EDUCATION

University of Georgia

Expected December 2026

Bachelor of Science in Computer Science

Systems-focused work in distributed computing, ML infrastructure, data engineering, and signal processing.

Georgia State University

Aug 2022 - Apr 2024

Prior coursework in Data Science